



Trends in co-fractionation mass spectrometry: A new gold-standard in global protein interaction network discovery

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Abstract

Co-fractionation mass spectrometry (CF-MS) uses biochemical fractionation to isolate and characterize macromolecular complexes from cellular lysates without the need for affinity tagging or capture. In recent years, this has emerged as a powerful technique for elucidating global protein-protein interaction networks in a wide variety of biospecimens. This review highlights the latest advancements in CF-MS experimental workflows including machine learning-guided analyses, for uncovering dynamic and high-resolution protein interaction landscapes with enhanced sensitivity, accuracy and throughput, enabling better biophysical characterization of endogenous protein complexes. By addressing challenges and emergent opportunities in the field, this review underscores the transformative potential of CF-MS in advancing our understanding of functional protein interaction networks in health and disease.

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Keywords

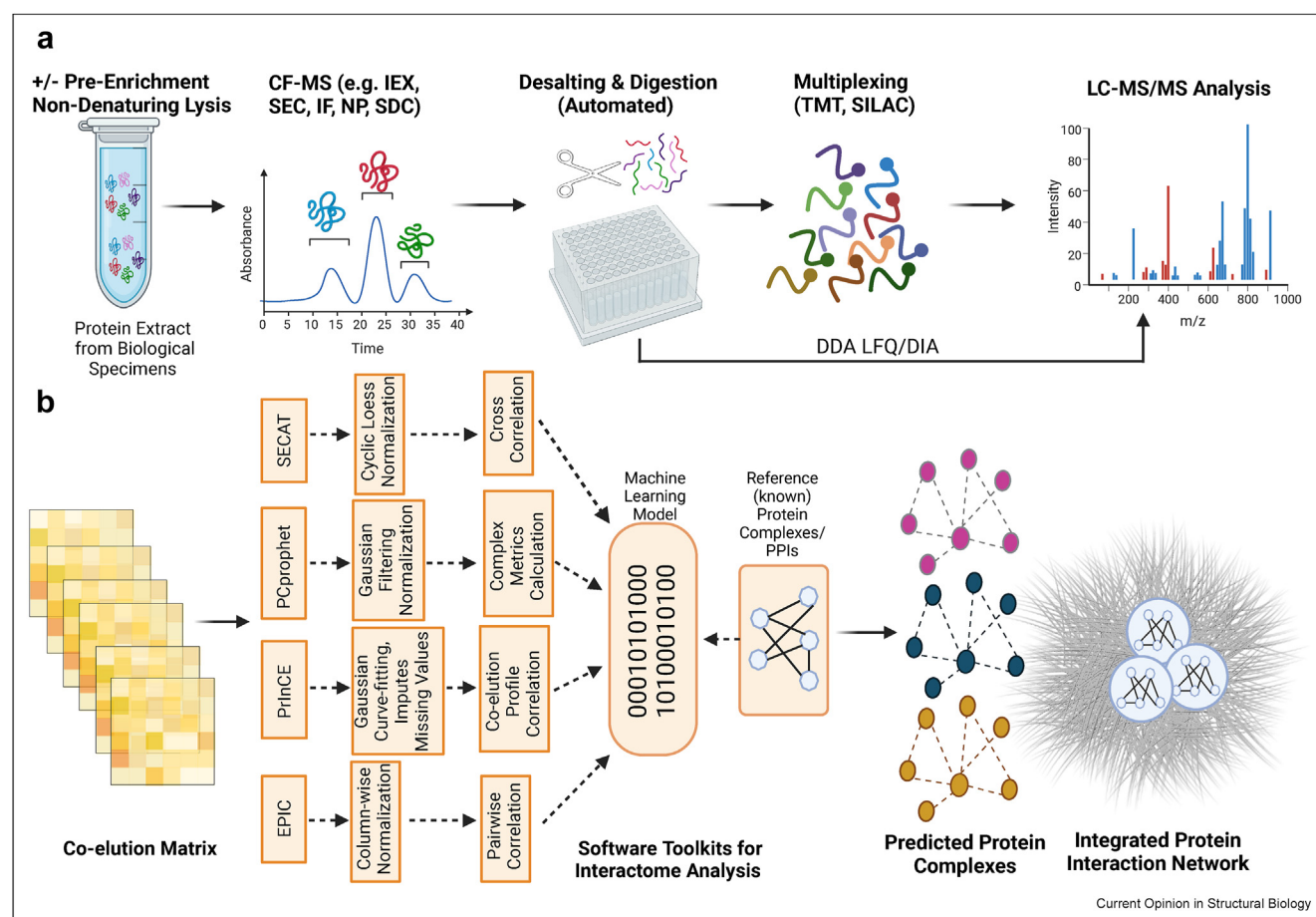
Co-fractionation mass spectrometry (CF-MS), Biochemical fractionation, High-throughput profiling, High performance liquid chromatography (HPLC), Protein-protein interaction (PPI), Protein interaction network (PIN), Protein complex, Data mining, Machine learning, TMT, SILAC, Label-free, DIA, DDA.

Introduction

Physical protein-protein interactions (PPIs) dynamically orchestrate cellular biochemistry and signal transduction modules necessary for physiological homeostasis [1,2]. The spatiotemporal organization of protein interaction networks (PINs) is governed by conserved biochemical cues, resulting in a dynamic assembly of functional protein complexes that drive cellular function [3,4]. These networks underlie virtually all normal cellular processes and adaptive responses, while their perturbation leads to disease states. An understanding of the composition and dynamics of these protein interaction networks has long attracted scientific attention and continues to represent a critical facet of research in human health and disease. Mass spectrometry (MS) has emerged as a cornerstone technology in the realm of proteomics, enabling the comprehensive analysis of complex protein mixtures. Traditional proteomic approaches like affinity-purification [5] or immunoprecipitation mass spectrometry [6] (AP-MS and IP-MS), while effective, grapple with low throughput and systematic challenges in studying global protein interaction networks. Co-fractionation mass spectrometry (CF-MS) has gained prominence for its capacity to unveil intricate native PINs in diverse pathobiological contexts and model systems in a less biased manner [7–9]. This innovative approach involves the integration of high-throughput chromatography with precision mass spectrometry, providing a powerful means to exploring the dynamic landscape of native protein complexes and associated cellular processes.

In the CF/MS technique, samples are lysed under gentle, non-denaturing condition [10]. Next, the soluble protein extracts are typically separated chromatographically followed by deep profiling of individual fractions by tandem mass spectrometry to reveal co-eluting protein interactors (Figure 1a). Fundamentally, CF-MS measures protein co-elution patterns to systematically infer protein assemblies. Data analysis is facilitated using machine learning tools that leverage the experimental patterns recorded for bona fide subunits or known (annotated) protein complexes as

Figure 1



A schematic illustration of the CF-MS workflow illustrating (a) newer experimental methodologies deployed including automated sample processing, multiplexed DDA and DIA modes of proteomic analyses, and (b) different computational tools, namely, EPIC, PrInCE, PCprophet and SECAT, enabling machine learning-guided PPI/protein complex discovery from DDA and DIA CF-MS datasets. IEX; Ion exchange, SEC; Size exclusion chromatography, SDC; Sucrose density centrifugation, IF; Isoelectric focusing, NP; Native PAGE.

orthogonal training information (prior knowledge/ experimentally validated protein interactions) (Figure 1b) [11–14]. While favoring the detection of stable multi-subunit assemblies, CF-MS captures a more comprehensive panoramic snapshot of protein interactions, which can precisely inform on the contextual physical rewiring of cells and tissues in an appropriately designed experiment.

In recent years, improved instrumentation, workflows and *in silico* analysis tools have reformed CF-MS capabilities, propelling forward the utility of this technology in increasingly diverse model systems and pathobiological contexts. The upcoming sections of this review navigate the innovative strategies deployed in CF-MS, ranging from recent advances in methodology/instrumentation to enhanced data analysis frameworks, that are significantly improving data scope and reliability, which are paving the way for democratizing unbiased global protein interactome discovery.

Methodological innovations: from bench to mass spectrometry

CF-MS is characteristically both instrument and labor-intensive, and has been generally viewed as technically nuanced. In the past few years, we and others have aimed at democratizing this core technology for community-wide uptake and utility by releasing a number of standardized and extensively benchmarked protocols in addition to original research articles [8,10,15–18]. Importantly, these methods describe the current state-of-the-art in CF-MS and illustrate newer performance benchmarks and advantages over older/alternate methods. The chronological summary provided in Table 1 provides an overview of notable past and recent CF-MS studies including key technical methods adopted. Despite these innovations, there remains little consensus on a universal CF-MS methodology. We reason that the choice of a suitable CF-MS method ultimately depends on the core objectives of the study and resource constraints, taking into consideration

Table 1

A chronological list of selected CF-MS research articles and methods/protocols. Relevant technical details and conceptual advancements are highlighted depicting the evolving state-of-the-art in the field over the years. DDA; Data-dependent acquisition, DIA; Data-independent acquisition, SEC: Size-exclusion chromatography, IEX; Ion-exchange chromatography, SDC; Sucrose density gradient centrifugation, IEF; Isoelectric focusing. N/A; not applicable in cases where information was not available.

References	Chromatographic Separation	#PPIs/Complexes	Biological material	CF-MS Data Analysis Tool	Highlights
Skinninger et al. [59]	SEC	5276/206	Honey bee midguts	PrInCE	Development of a harmonized PPI resource (CFdb) via meta-analysis of 206 CF-MS datasets. Prediction of honeybee midgut interactome via DIA SEC CF-MS.
Zilocchi et al. [8]	IEX, SEC	20,742/543	Mouse brain tissue and human embryonal carcinoma cells (NTERA-2)	MACP; formally packaged CF-MS toolkit based on a previous study [47]	Crosslinked DDA CF-MS protocol for isolating mitochondrial protein complexes. Development of the MACP toolkit for CF-MS data analysis.
Akinlaja et al. [10]	SEC	–	–	PrInCE	Comprehensive CF-MS protocol for label-free DDA proteomics
Bokor et al. [14]	SEC	–	–	SECAT	Stepwise protocol for analyzing DIA CF-MS (SEC) datasets using the SECAT toolkit.
Bludau et al. [24]	SEC	(N/A)/644	Human monocyte cell line, THP-1	CCprofiler	Optimization of a short LC-MS gradient in SWATH DIA mode for accelerated CF-MS data acquisition.
Gaun et al. [29]	SEC	–	Plasma	Customized CF-MS data analysis pipeline	Comprehensive CF-MS sample preparation protocol for analysis of blood plasma protein complexes. Automated sample preparation steps for DDA label-free and TMT multiplexed analysis.
Sae-Lee et al. [17]	IEX, SEC & HIC	3229/100	Primary human RBCs	Customized CF-MS data analysis pipeline	Prediction of human RBC interactome via multidimensional DDA CF-MS.
Xu et al. [33]	IEX, SEC & SDC	24,092/291	Synechocystis	EPIC	Multi-dimensional DDA CF-MS revealing protein complexes in Synechocystis.
Havugimana, Goel et al. [15]	IEC	25,235/772	Human mammary tissue-derived cell lines (MCF10A, MCF7 and MDA-MB-231)	EPIC	TMT multiplexed CF-MS workflow integrating automated sample processing for efficient, accelerated and accurate PPI discovery. Generated a large breast cancer interaction landscape.
Schlossarek et al. [60]	–	–	–	PROMISed	Development of a CF-MS data analysis toolkit.
Skinninger et al. [54]	–	–	–	PrInCE	Development of a CF-MS data analysis toolkit.
Skinninger et al. [61]	–	47,575/2386	Human and mouse material	PrInCE	Built a draft human interactome and predicted 700,000 PPIs across 27 eukaryotic species via meta-

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Table 1. (continued)					
References	Chromatographic Separation	#PPIs/Complexes	Biological material	CF-MS Data Analysis Tool	Highlights
McWhite et al. [7]	–	–	–	cfmsflow	analysis of 206 CF-MS experiments. CF-MS sample preparation protocol for low vs high-salt samples. Development of a CF-MS data analysis pipeline, <i>cfmsflow</i> .
Lee et al. [62]	SEC	(N/A)/246	<i>Oryza sativa</i> (Rice), <i>Glycine max</i> (Fabales) <i>Gossypium hirsutum</i> (cotton) and <i>Arabidopsis thaliana</i>	Customized CF-MS data analysis pipeline.	Prediction of homo-heteromeric protein complexes that drive neofunctionalization, by DDA CF-MS.
Fossati et al. [13]	SEC, IEX	(N/A)/900	HeLa CCL2 cell line	PCprophet	Development of a CF-MS data analysis toolkit. Benchmarked PCprophet performance against CCprofiler and EPIC.
Swapna et al. [63]	IEX	2063/93	<i>Toxoplasma gondii</i>	Customized CF-MS data analysis pipeline based on EPIC [12]	Predicted protein complexes in T. gondii by DDA CF-MS.
Lee et al. [34]	SEC, IEX	(N/A)/770	<i>Oryza sativa</i> (Rice) aleurone-subaleurone	Customized CF-MS data analysis pipeline.	Predicted protein complexes by multidimensional DDA CF-MS.
Rosenberger et al. [64]	SEC	1068/(N/A)	HeLa CCL2 [50]	SECAT	Development of a CF-MS data analysis toolkit.
Pourhaghighi et al. [28]	IEX, IEF	27,403/1000		EPIC	Predicted murine brain protein complexes by multidimensional DDA CF-MS.
Heusel et al. [50]	SEC	(N/A)/949	HeLa CCL2 cell line	CCprofiler	Predicted protein complexes by CF-MS using SWATH DIA proteomics.
McWhite et al. [18]	SEC, IEX & IEF	886/117	13 plant species	Customized CF-MS data analysis pipeline	Predicted protein complexes across 13 plant species by multidimensional DDA CF-MS.
Hu et al. [12]	IEX	16,098/612	<i>Caenorhabditis elegans</i>	EPIC	Development of a CF-MS data analysis toolkit. Benchmarked performance against PrInCE.
Heusel et al. [49]	SEC	(N/A)/621	HEK293 cell line	CCprofiler	Development of a CF-MS data analysis toolkit. First SWATH DIA proteomics method in CF-MS.
Moutaoufik et al. [47]	SEC, IEX	3086/139	NTERA2 and differentiated neuronal-like cells.	Customized CF-MS data analysis pipeline.	Predicted mitochondrial protein complexes by multidimensional crosslinked DDA CF-MS.
Gilbert et al. [65]	SEC	73,929/963	<i>Arabidopsis thaliana</i> roots	Customized CF-MS data analysis pipeline.	Predicted membrane protein complexes by DDA CF-MS.

whether the goal/need is to: (i) identify new protein complexes versus studying known complexes, (ii) elucidating protein interaction dynamics across a sub-cellular milieu such as specific organellar or cellular compartments including cytoplasm [19], nucleus [19] and membrane [20,21], necessitating prior subcellular enrichment and, (iii) resource limitations, where instrument availability, sample quantities and/or project timeline are constraining. That said, the availability of newer experimental reagents, advancements in instrument performance and exponential progress in machine-learning/data-analysis capabilities are collectively expected to consolidate CF-MS workflows into a seamless turnkey approach.

A major determinant of successful outcomes is resolving protein assemblies prior to detection by LC/MS, which is challenging given the highly complex nature of biological mixtures. Both ion-exchange (IEX) and size exclusion chromatography (SEC) are well-characterized biochemical co-fractionation approaches frequently described in the CF-MS literature [18,19,22–25]. Other orthogonal methods such as native-gel separation [26], density gradient centrifugation [19,27] and isoelectric focusing [19,28] have also been explored. But these methods are inherently low throughput/low-resolution in nature and entail inconvenient downstream sample processing steps (e.g. protein extraction from native gel slices), making them somewhat less impactful to date. Compared to SEC, ion-exchange offers appreciably higher resolution separation of protein complexes based on net charge but is potentially agnostic to high-salt-sensitive complexes [29]. Co-assembling anion and cation exchange matrices further boosts the resolving power of IEX HPLC [17,19]. SEC on the other hand resolves complexes based on molecular weight and has been more frequently utilized in CF-MS, partly because SEC is essentially a fundamental adaptation of the more traditional molecular weight-based native gel separation technique which has historically dominated biochemical studies exploring complexosomes [26,30–32]. Also, given the limited dynamic resolution of SEC, handling fewer co-eluting protein fractions is less burdensome for bottom-up proteomics analysis, therefore attracting user preference.

We and others originally implemented IEX chromatography to address unmet challenges in SEC CF-MS, mainly: (i) low dynamic range of separation, biasing detection towards high-abundance protein complexes and (ii) chance co-elution of functionally unrelated proteins, prompting false positive PPIs [17,19,33]. IEX largely addresses these issues by allowing more extensive fractionation of protein mixtures using shallow salt gradients. Various groups have even proposed combining IEX with SEC to widen orthogonal separation and diminish chance co-elution [17,18,34]. For instance, Lee *et al.* reported that this combinatorial separation

strategy led to at least 70% proteins falling within just two to four co-elution peaks, suggesting sharper resolution [34]. A notable downside to this joint approach, however, is the higher number of co-eluting protein fractions which substantially burdens manual sample processing, consequently introducing inconsistencies and raising error rates. The inclusion of multiple experimental conditions and replicates can further amplify noise. Sample clean-up is particularly critical and can be manually cumbersome when dealing with a large number of CF-MS fractions. Improperly desalted samples can cause poor chromatographic resolution, capillary clogging and ion suppression [35,36], effectively penalizing protein and interactome coverage. Furthermore, label-free data-dependent acquisition constrains instrument time and carries potentially higher qualitative and quantitative variability which impose significant technical bottlenecks in quality PPI discovery, particularly for pan-interactome studies. In fact, some of the early pioneering work on CF/MS utilizing label-free quantitation, profiled over 500 fractions to achieve high resolution [19]. Despite enabling a deep interactome coverage spanning >600 protein complexes (PMID), the enormous number of samples and data acquisition time needed in these studies have posed a formidable challenge in routine and universal implementation of CF/MS.

To this end, multiplexing strategies, using isotopic SILAC [37]/SILAM [16] and more lately using isobaric tandem mass tags (TMT) [15,28,29,38,39], have been adopted in CF-MS which now permit unprecedented multiplexing of up to 18 unique samples [29], increasing throughput and quantitative rigor. SILAC/SILAM may be advantageous if fewer experimental conditions are desired, particularly since MS1-based isotopic protein quantification is practically unaffected by ratio distortion which is a well-documented caveat in MS2-based isobaric labelling. An exemplary *in vivo* CF-MS endeavor incorporating SILAM led to the identification of 125,696 PPIs across seven mouse organs, representing the largest murine PPI cohort reported to date [16]. In comparison, isobaric barcoding offers more versatility, and distorted quantification has been widely addressed over the years with dramatically improved MS1/MS2 resolution and narrower isolation widths to mitigate peptide co-fragmentation [40]. In addition to harnessing the wider multiplexing capabilities of isobaric/TMT tags, automated sample processing, originally described for phosphoproteomics [41] and AP-MS [42], has also been implemented in recent CF-MS studies [15,29]. Automated sample clean-up using magnetic carboxylic beads combined with TMT-multiplexing was shown to significantly increase efficiency, quantitative reproducibility and PPI coverage in a recent work by our group [15]. This multiplexed CF/MS (mCF/MS) method produced a large breast cancer interaction network comprising 772 putative protein

complexes from multibed IEX chromatography. Notably, here isobaric multiplexing alone reduced MS instrument time to 16 days from 96 days [15]. Another useful CF-MS method describing the extraction of plasma complexes by SEC echoed the immense utility of automated sample-preparation and TMT-multiplexing [29] in increasing efficiency and throughput.

Generally, preserving protein complex integrity throughout sample processing, including the identification of transient/weak interactors is challenging. *In vivo* chemical cross-linking of proteins is a viable approach that has been adapted in recent years to improve the identification of cellular complexes by CF-MS and other approaches [8,43–47].

The markedly improved scanning speed and sensitivity of newer generation mass spectrometers has sparked renewed interest in data-independent acquisition (DIA), which holds great promise for CF-MS. Unbiased precursor ion selection and fragmentation via DIA ensures; (i) better sampling of low-abundance ions resulting in minimal missing values and hence a more “complete” PPI network, and (ii) comparably higher rigor than classical stochastic DDA-driven approaches, extending the discovery of high-confidence PPIs [48,49]. Three relevant studies have laid the founding stone for integrating CF-MS with DIA mass spectrometry [24,49,50]. An important consideration in DIA relates to the generation of sample-specific spectral-libraries which favorably enhance protein and PPI coverage, as noted by Hay et al. and others [48,51]. Such sample-specific spectral libraries ensure accuracy in protein ID assignment which consequently limits false positive PPI predictions [48,51]. Exemplar work by Heusel et al. and Bludau et al. describe the use of sample-specific reference peptide spectrum libraries generated first from DDA-MS to subsequently query SEC co-fractionation data by DIA-based sequential window acquisition of all theoretical fragment ion spectra mass spectrometry (SWATH) [49,50]. This approach led Heusel et al. to identify 949 complexes in HeLa cells and attain a high degree of reproducibility in mapping remodeled physical protein interaction networks across interphase and mitosis [50]. To further accelerate data acquisition by DIA, Bludau et al. incorporated short gradient chromatography, notably using a 21-min gradient (instead of ~60 min-gradient described previously) and reported 644 high confidence complexes including 17 rewired complexes in differentiating THP1 cells [24]. Under otherwise similar experimental parameters, this represented an order of magnitude reduction in data acquisition time. Comprehensive SEC-SWATH-MS protocols by this group outlines SEC sample processing and data analysis using the complex-centric profiler (CCprofiler) machine learning suite [52,53].

Collectively, automated sample processing is tremendously beneficial to CF-MS workflows as it cumulatively

reduces both “hands-on” time and human error. Multiplexed DDA offers more accelerated data acquisition even under standard 60–120-min chromatographic gradients while boosting signal from low-abundance ions. But overall DIA is potentially a better and viable alternative to DDA in that it more adequately addresses stochasticity and low-abundance-related caveats. As the field progresses, it is expected that the dilemma of using sample-specific DDA spectral libraries that entail significant time and resource investment or using pursuing predictive/*in silico*-established libraries that limit reliable protein identifications, will be appropriately reconciled. While certain groups are already exploring SEC DIA [24,50], coupling higher resolution IEX with DIA also appears to be a promising parallel direction in CF-MS.

Machine learning tools for predicting protein interaction networks from CF-MS data

A number of open-access *in silico* tools have been developed to confidently infer protein interactions from DDA or DIA CF-MS data (Table 1). These include PrInCE [54], EPIC [12], CCprofiler [52], SECAT [14] and PCprophet [13], which systematically quantify the abundance and interaction stoichiometry of protein complexes. Whereas PrInCE and EPIC were established to handle DDA CF-MS datasets, CCprofiler, SECAT and PCprophet were designed to primarily analyze DIA data. Notably, the more recently developed, PCprophet was benchmarked rigorously on both DDA (label-free, SILAC) and DIA CF-MS datasets. These tools utilize a protein co-elution profile as input data, making them versatile. Figure 1b illustrates the general computational scheme for PPI/protein complex prediction. All five state-of-the-art tools fundamentally rely on (i) a “guilt-by-association” model [9] which necessitates that physically interacting proteins co-elute together, and (ii) a “reference” set of evolutionarily conserved curated protein complexes or binary PPIs [16]. The latter is central to the semi-supervised machine learning-guided modality of PPI/complex prediction by these tools, as discussed in the subsections below. CORUM, IntAct, GO, iRefIndex, hu.MAP, PrePPI and STRING are among the gold-standard repositories from which reference protein interactions are sourced for both training and benchmarking [12,14,19]. Randomized PPIs/complexes and other functional evidence like gene co-expression and/or gene ontology (GO) can enhance predicted protein interactions. In the following sections, we highlight some distinguishing facets of each tool to illustrate user applicability and summarize these in Table 2.

1. PrInCE and EPIC: the original stalwarts of DDA CF-MS data analysis

PrInCE [54] and EPIC [12] were the first bioinformatics tools published for handling DDA CF-MS datasets. Both

Table 2

Comparative illustration of some of the key features of the prominent CF-MS computational tools, EPIC, PrInCE, ccProfiler, SECAT and PCprophet.

Feature	EPIC	PrInCE	CCprofiler	SECAT	PCprophet
Input dataset type (benchmarked/tested on)	DDA; SEC, IEX, SDC and IEF (TMT, SILAC and label-free)	DDA; SEC and IEX (SILAC and label-free)	DIA; SEC	DIA; SEC	DDA and DIA; SEC, IEX (SILAC and label-free)
Data imputation	–	Mean of 5 neighboring data points	Spline-fit function to impute missing values flanked by at least 2 data points	–	Average of 2 neighboring fractions
Normalization	Column-wise normalization	Gaussian curve fitting (peak dimensions)	Normalize signal to a spike-in standard	Cyclic Loess	Gaussian filtering and one-dimension linear interpolation
Metrics used for generating probabilistic interactions	Euclidean, Jaccard, Apex, Pearson, Pearson with Poisson noise, weighted cross correlation, mutual information, and Bayes correlation	One minus Pearson's correlation coefficient, the associated correlation p-value, Euclidean distance, peak location and co–apex score	Protein complex hypothesis generated by referencing against CORUM or BioPlex and STRING	Cross-correlation-based scores, Monomer-based scores, Maximal and total information coefficient-based scores, SEC profile intersection-based scores, Protein-abundance-based scores, Kickstart score	Peak-picking and Distance-based clustering
Machine learning model	Random Forest or Support Vector Machine	Naïve Bayes Classifier	–	XGBoost classifier	Random Forest classifier
Reference database(s) used for training	CORUM, IntAct and GO	CORUM	CORUM, BioPlex and STRING	CORUM or STRING	CORUM
Optimization of PPI/complex predictions	Nested 2-fold cross validation against curated complexes. Benchmark predictions using composite score	Ten-fold cross-validation	Estimate assembled mass fraction and feature mass fraction. p-value estimation and multiple testing by Benjamini- Hochberg	Ten-fold cross-validation. Storey-Tibshirani q-value estimation	Five-fold cross-validation. GO term score filtering and FDR estimation
PPI/complex differential abundance quantification	–	Paired t-test	beta-regression model and p-value estimation. Multiple testing correction by Benjamini-Hochberg adjustment	ProteoVIPER algorithm	Bayesian inference method
Module for building PPI network	ClusterONE	ClusterONE	–	Custom network-centric data integration. Test statistics and ProteoVIPER enrichment scores integrated by Empirical Brown's method to produce composite protein complex metrics	NetworkX

tools enable the retrieval of curated co-complex information and the prediction of novel complex assemblies from experimental data. The EPIC toolkit traces its origins to the first multidimensional CF-MS study that charted soluble protein complexes in human cell lines [19]. Modular algorithms from this study were later formally packaged into a dedicated and publicly available software [12]. For pre-processing, EPIC first eliminates proteins quantified in only one fraction and performs column-wise normalization (MS1/MS2 intensities divided by the number of proteins in corresponding fractions). It then computes pairwise correlation from the normalized co-elution matrix by using 8 different distance metrics (Euclidean, Jaccard, Apex, Pearson, Pearson with Poisson noise, weighted cross correlation, mutual information, and Bayes correlation). A random forest or support vector machine classifier generates probabilistic PPIs by training on reference true and randomized decoy interaction datasets based on CORUM, IntAct and GO, as a means of two-fold cross-validation of PPI predictions. We find that integration of functional evidence (gene co-expression, co-citation etc.) may contextually boost PPI predictions [13,19,28] and is likely more helpful when the experimental datasets are sparse. Accordingly, we showed that deep and reproducible multiplexed experimental data from IEX co-fractionation alone could generate accurate macromolecular predictions. Integrating additional functional evidence from GeneMania/STRING only slightly improved performance [15]. Additionally, we and others have shown that the random forest classifier outperforms support vector machine in overall predictions including improved recall and accuracy [12,13]. The predicted binary protein interactions are assembled into protein complexes using the ClusterOne partitioning algorithm [55]. EPIC has been benchmarked on label-free, TMT and SILAC datasets derived from IEX, SEC, iso-electric focusing and sucrose density gradient centrifugation-based co-fractionation methods.

PrInCE too was developed as an efficient CF-MS interactome analysis pipeline incorporating conceptual advancements over previous bioinformatics methods published from the same group [37]. In contrast to EPIC, PrInCE employs conservative pre-processing steps. It utilizes a Gauss-Build function to model Gaussian curves, limited to those with data in at least five consecutive fractions. Replicate elution profiles are corrected to align peak positions, unlike EPIC's direct column-wise normalization of MS1 or MS2 intensities. Additionally, PrInCE imputes missing values in co-fractionation profiles by averaging five neighboring data points, setting remaining missing values to zero. In contrast, EPIC performs no such imputation. Furthermore, PrInCE conducts differential abundance calculations using paired t-test for co-eluting protein pairs, while EPIC lacks built-in functions for such analysis, requiring separate procedures as described recently [15,38]. Like EPIC, PrInCE utilizes diverse distance metrics for co-elution

profile correlation and applies a naïve Bayes classifier trained on CORUM reference complexes to predict binary protein interactions. As with EPIC, high scoring PPIs are organized into protein complexes by ClusterOne [55]. In alignment with its intended use, PrInCE has been successfully implemented on both SILAC and label-free DDA CF-MS datasets.

By benchmarking with SILAC data, PrInCE importantly demonstrated that protein abundance correlated well with predicted protein interactors [11]. Despite EPIC's inherently underestimates the utility of protein quantitation in ascribing confidence to protein interactions, using a common SILAC CF-MS dataset EPIC still produced a higher composite score [12] signifying higher overall prediction accuracy relative to PrInCE. Like PrInCE and other CF-MS pipelines described below, in recent years we have pivoted to incorporating protein quantitation information for signifying differential protein complexes predicted using EPIC [15,38].

2. Coming of age with DIA CF-MS: CCprofiler, SECAT and PCprophet

CCprofiler [49], SECAT [14] and PCprophet [13] are all based on SWATH DIA proteomics, whose selective advantages over other DIA strategies have been reviewed before [56]. Unlike the interaction-centric algorithms of EPIC and PrInCE that infer protein complexes from predicted intermediate binary PPIs, CCprofiler extends a complex-centric analysis to the experimental data producing directly protein complex predictions [49,53]. This has been rationalized as an advantage over ClusterOne which potentially amplifies network-level noise [13]. CCprofiler imports precursor/MS1-level data, normalized to a spike-in standard, and imputes missing values using background signals from adjacent fractions. Quantitative protein elution events are identified through peak detection, summing peptide signals per parent protein across replicate runs. Protein complex hypotheses are generated by referencing CORUM, BioPlex, and STRING entries, with error estimation using target-decoy entries from curated databases. Subsequently, CCprofiler integrated differential protein abundance calculations to differentiate protein mass redistribution within specific assemblies from compositional changes [50]. Notably, since CCprofiler lacks a machine-learning component, it basically only recalls existing well-characterized complexes from curated databases and does not discover novel protein assemblies. This imposes a bias towards studying known protein complexes in new experimental datasets where the output is proportionally limited to the size of the reference dataset.

To overcome this constraint, SECAT [14] was developed around a network-centric strategy generating

machine learning-driven probabilistic PPI networks instead of explicitly inferring protein complexes by similarity to reference databases, as with CCprofiler. SECAT preprocesses data differently, normalizing peptide-level quantitation with cyclic loess and optionally applying protein-level peak picking or detrending to define peak boundaries. Notably, default exclusion of peak picking/detrending in SECAT was found to enhance mean PPI quantification in benchmarking against previous datasets [50], suggesting minimal data preprocessing requirements and hence better computational efficiency over other tools. Nevertheless, depending on the SEC data type, peak picking and/or detrending can potentially benefit PPI scoring [14]. Candidate PPI pairs are defined by computing a series of 11 partial scores that aid in preliminary discrimination of true positive PPIs. An XGBoost machine learning classifier is trained on both reference positive and negative decoy PPIs derived from CORUM or other sources (STRING/PrePPI) to iteratively refine PPI predictions. Like CCprofiler, summed peptide-level intensities are log2-transformed for differential subunit abundance analyses between the experimental conditions. Moreover, like EPIC, SECAT conceptually performs cross validation using a hold-out set of PPIs. Statistical validation is computed by a q-value framework as with CCprofiler, and akin to the precision-recall metric used in EPIC and PrInCE. Given the basic difference in the functionalities of SECAT and CCprofiler (complex-versus network-centric), an objective comparison of outputs was difficult to achieve. But gene set enrichment analysis of interactors inferred by SEC CF-MS of interphase- and mitotically-arrested HeLa cell extracts demonstrated greater enrichment of cell-cycle related functional terms by SECAT relative to CCprofiler, suggesting better systems-level interpretation. An elaborate and helpful stepwise protocol on implementing SECAT for user-specific datasets has been reported [14].

The third toolkit, PCprophet [13], like CCprofiler infers protein complexes directly from co-fractionation data. But unlike CCprofiler, protein complex predictions here are guided by the random forest machine-learning classifier and uniquely thresholded by error-rate estimation based on pairwise subunit GO term semantic similarity, thereby enabling robust discovery of novel protein complexes. PCprophet builds upon the rigorous frameworks of SECAT, PrInCE, and EPIC, extending multiparametric data processing. Gaussian filtering is applied to denoise raw elution profiles in preprocessing, with missing values imputed by averaging neighboring fractions and curve fitting enabled by linear interpolation. Candidate complexes are defined based on apex intensities of fraction peaks and distance-based linkage clustering. PCprophet calculates four additional co-elution profile metrics for candidate complexes: average intensity difference, average local correlation,

average shift of apex fraction, and average full width at half maximum (FWHM) for associated protein peaks. It utilizes random forest over a reference dataset from CORUM or PPI sources, and benchmarking using FDR estimated from GO semantic similarity [57] scores (cellular component, molecular function and biological process) of complex subunits. This mechanism of GO-derived FDR control can help deduce functional significance for novel PPIs or complexes [13]. Protein complex size can further be iteratively collapsed through one of four modes based on either; (i) GO score, (ii) difference between apparent molecular weight by SEC and extrapolated molecular weight from a calibration curve, (iii) probability derived via machine-learning and (iv) number of subunits per complex. A Bayesian inference method translates protein abundance metrics into protein interaction likelihood across conditions while penalizing proteins with high variability. Importantly, objective benchmarking of PCprophet's performance against EPIC (random forest) and CCprofiler on both DIA [50] and DDA-SILAC [37] CF-MS datasets demonstrated superior recall of complexes and PPIs from CORUM and STRING/BioPlex, respectively. Additionally, PCprophet produced more accurate complex size similarities for CORUM complexes than EPIC, suggesting better overall performance. However, a potential limitation arises from the need to downscale input co-elution datasets to a maximum of 72 fractions during normalization. This constraint might affect protein assembly predictions from highly orthogonal IEX CF-MS datasets and requires evaluation.

With the exception of EPIC, most CF-MS analysis tools described have been mainly benchmarked on low-resolution SEC co-fractionation datasets. Handling substantially larger data matrices, generated from extensive IEX co-fractionation, is computationally intensive and it remains to be determined how well some of these tools estimate precision-recall at the PPI/complex level.

Concluding remarks

Protein interactions play an important role in the maintenance of cellular and physiological homeostasis. The ability to perturb disease-causing protein interactions has historically attracted widespread clinical interest. But lower resolution approaches have posed a major bottleneck to these efforts, leading to fewer candidate therapies progressing to clinical trials [58]. Co-fractionation mass spectrometry is an actively evolving area of functional proteomics that can better elucidate physical protein interactions implicated in health and disease. Unlike AP/IP-MS or traditional proximity labelling approaches, CF-MS is advantageous in that it allows for elucidating native protein interactions all in a single experiment. However, furthering the resolution by incorporating orthogonal co-fractionation strategies is important particularly for the detection of low-abundant

complexes, as is a unified effort to build customized computational solutions for CF-MS data analyses. Still, key technical reforms to CF-MS methodology and data analyses pipelines over the past few years have increasingly simplified user applicability and enhanced robustness of protein interaction predictions. Looking forward, we anticipate that sustained improvements in instrumentation and computational resources will pave way for this gold-standard proteomic technology to unravel a large repertoire of clinically significant targets that are yet untapped.

Author contributions

R.K.G. researched the literature, conceived and prepared the original draft of the manuscript. N.B. prepared the figure and table under the supervision of R.K.G. All authors edited the manuscript. A.E. mentors R.K.G. and N.B.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

No data was used for the research described in the article.

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